# AAE4009 – Lab 1

Cognitive State Detection in Pilots Using Machine Learning

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## 1 Introduction

Aviation accidents today are increasingly caused by human error rather than mechanical failure. A key contributor is the loss of situational awareness, where pilots become distracted, fatigued, or cognitively overloaded without any external warning. Monitoring pilot cognitive states in real time could allow safety systems to intervene before an error occurs.

The brain regions involved in attention and stress response are well established. Frontal cortex areas (Brodmann areas 8, 9, 46) govern executive attention and are active during sustained focus; temporal and parietal regions mediate multi-task coordination; limbic structures drive acute stress responses. EEG measures the oscillatory electrical activity of these regions at millisecond resolution, making it the most direct sensor for cognitive state inference. Different states produce characteristic patterns across standard frequency bands: theta (4–8 Hz) increases during cognitive load, alpha (8–13 Hz) suppresses during active engagement, and beta (13–30 Hz) is associated with alertness and attention switching.

The objective of this project is to build machine learning classifiers capable of detecting three pilot cognitive states - Channelized Attention (CA), Diverted Attention (DA), and Startle/Surprise (SS) - from EEG and physiological signals. A key focus beyond raw accuracy is whether models can generalise to pilots they were not trained on, which is the condition any real deployed system must satisfy.

## 2 Research Methodologies

### 2.1 Dataset

The dataset contains physiological recordings from pilots subjected to three controlled cognitive state induction protocols. CA was induced via an engaging puzzle game; DA via a monitoring task with periodic arithmetic interruptions; SS via film clips with jump scares. Each experiment contains one target state interleaved with a baseline (no-event) condition. Events are labelled A (baseline), B (SS), C (CA), and D (DA). A simplified set of 6,000 rows covers 4 pilots; an intermediate set covers 1 million lines of the same 4 pilots but at more frequent intervals; finally, an extended dataset of approximately 4 million rows covers 18 pilots sampled at 256 Hz.

### 2.2 Pre-processing

Pilot identity was defined as pilot $= 100 \times \text{seat} + \text{crew}$, since neither column alone is a unique identifier. Per-pilot MinMax normalisation was applied before splitting to try to prevent inter-pilot amplitude differences from affecting model performance. The training and test sets were initially constructed using a stratified shuffle split (70/30) that preserves class proportions across the split. Later, to ensure more robust results and prevent data leakage, a cross-validation scheme was implemented where each pilot acted as one fold (Leave One Group Out cross-validation).

### 2.3 Feature Engineering

Raw EEG amplitude at a single timestep carries almost no class separability, confirmed by ANOVA (Analysis of Variance). The primary feature engineering step transforms raw signals into frequency-domain band power features using Welch's power spectral density method. For each 256-sample sliding window (1 second/time unit at 256 Hz), the power in four EEG frequency bands is computed per channel: theta, alpha,

beta, and gamma. The delta band was excluded after diagnostic analysis revealed all-zero values due to insufficient frequency resolution at the chosen FFT parameters. Physiological signals (ECG, respiration rate $r$, GSR) were summarised per window as mean, standard deviation, and inter-window rate of change. This reduced the dataset from approximately 4 million time series rows to approximately 75,000 windows of 89 features.

### 2.4 Statistical Analysis

ANOVA (Analysis of Variance) is a statistical test that measures whether the mean of a variable differs significantly across groups - in this case, used to quantify how much of the variance in each EEG feature can be explained by cognitive state class membership, expressed as the effect size $\eta^2$. One-way ANOVA was used to measure how discriminative each feature is with respect to cognitive state class. The effect size $\eta^2$ is computed as:

$$\eta^2 = \frac{SS_{\text{between}}}{SS_{\text{total}}} \tag{1}$$

where values of 0.06 and 0.14 correspond to medium and large effects respectively. This was applied to both raw and transformed features across both the 4-pilot and 18-pilot datasets to assess how feature separability scales with population diversity.

### 2.5 Models and Evaluation

Six classifiers were evaluated on the simplified dataset: KNN, Naive Bayes, Decision Tree, linear SVM, MLP, and Gradient Boosting. On the transformed 18-pilot dataset, LightGBM, XGBoost, and Random Forest were added. Models were also tested with a restricted beta/gamma ($\beta/\gamma$) feature set, motivated by between-pilot variability analysis showing that theta and alpha features encode pilot identity more strongly than cognitive state.

Two evaluation schemes were deliberately contrasted. The *within-session* stratified split tests performance on data from pilots already seen during training. *Pilot-out cross-validation* holds one complete pilot's data as the test set and trains on all remaining pilots - this is the operationally relevant test. The primary metric throughout is macro-averaged F1, which weights all four classes (events A, B, C and D) equally regardless of frequency, ensuring sensitivity to the rare but safety-critical DA and SS states.

## 3 Results

### 3.1 Data Structure and Class Distribution

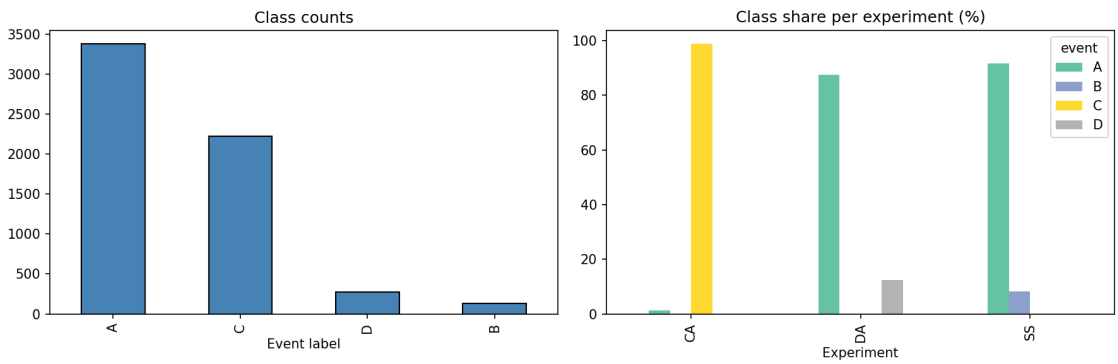


Figure 1: Class counts (left) and class share per experiment (right). Note that DA and SS comprise only 8–13% of their respective sessions.

Figure 1 reveals severe class imbalance: a model that basically acts as a binary classifier between CA and baseline would have above 85 % accuracy, while completely failing to detect any dangerous cognitive state event. This makes accuracy a misleading primary metric.

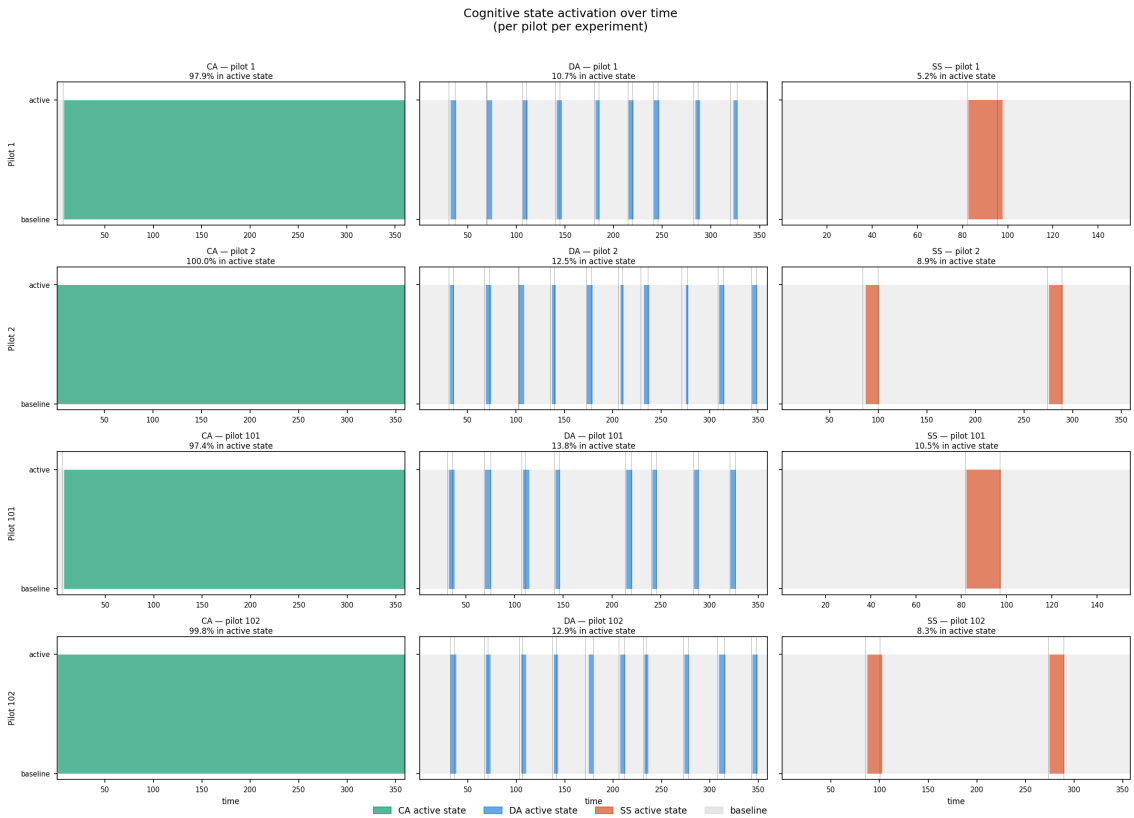


Figure 2: Cognitive state activation over time per pilot per experiment (small dataset). CA is sustained for nearly the entire session; DA appears as repeated short pulses; SS occurs in 1 or 2 discrete blocks per pilot.

Figure 2 shows that each cognitive state has a distinct temporal structure. DA detection is especially challenging because events are brief — the classifier must catch a short signal transition rather than a sustained pattern.

## 3.2 Feature Separability

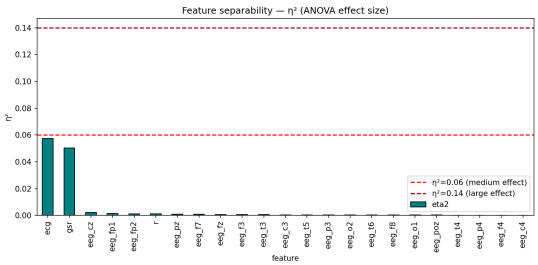


Figure 3: $\eta^2$ for raw time-domain features. Only ECG and GSR show meaningful separability; all EEG channels are near zero.

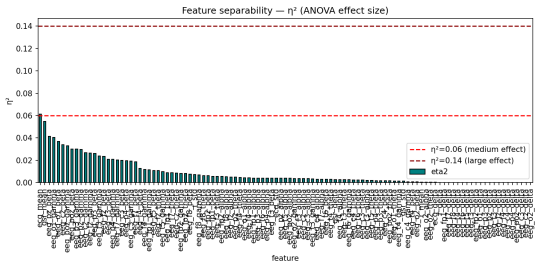


Figure 4: $\eta^2$ for frequency-domain features. Beta and gamma band EEG channels emerge as the most discriminative after Welch transformation.

Figures 3 and 4 show the effect of the frequency transformation. Raw EEG amplitude is essentially non-discriminative; beta and gamma band powers are moderately useful. ECG and GSR remain the top features, consistent with their role as direct physiological stress markers.

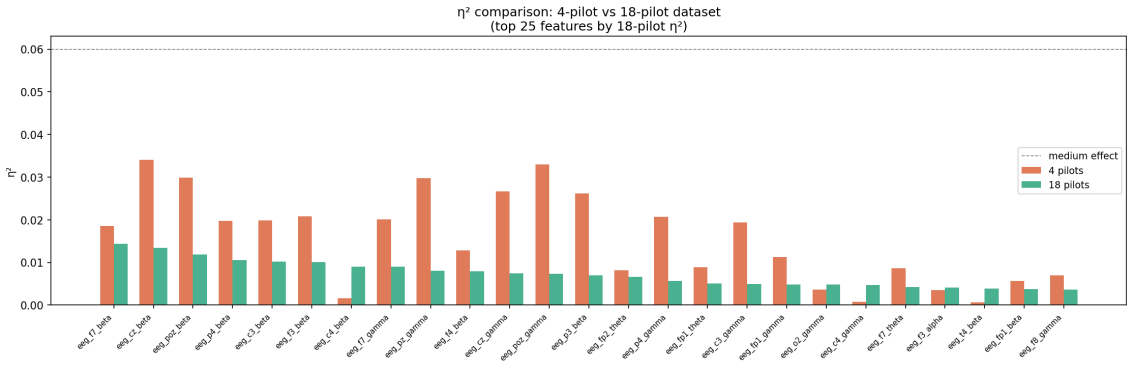


Figure 5: $\eta^2$ comparison between the 4-pilot and 18-pilot datasets for the top 25 transformed features. Effect sizes consistently halve or worse when the population increases.

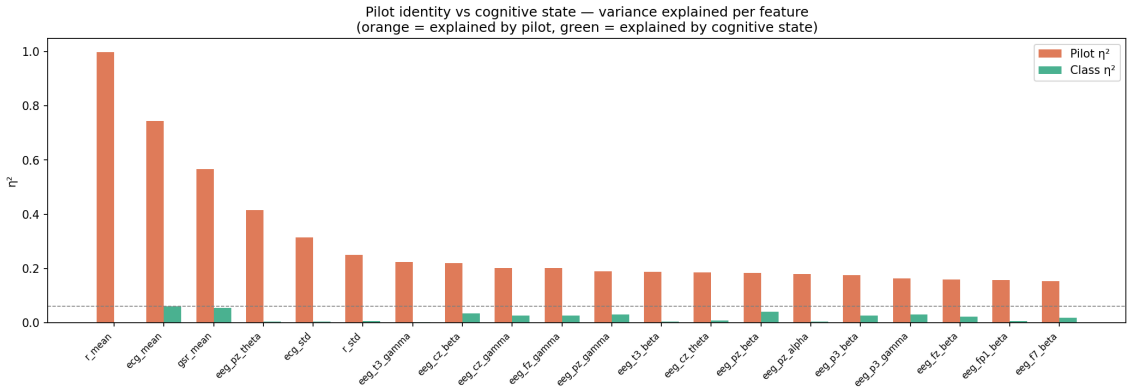


Figure 6: Pilot identity $\eta^2$ versus cognitive state $\eta^2$ per feature. Pilot identity explains more variance in EEG band powers than cognitive state does for most features.

Figures 5 and 6 together identify the root cause of poor cross-subject performance: separability halves as the population grows, and the features that should encode cognitive state are primarily encoding individual pilot identity instead.

## 3.3 Model Performance

Table 1 reports the full benchmark results on the simplified 4-pilot dataset, including macro-averaged F1 alongside accuracy, precision, and recall. Because baseline events (A and C) account for over 85% of the data, a model can achieve high accuracy simply by predicting baseline or Channelized Attention (CA) most of the time while ignoring the cognitive states that matter. The contrast between test accuracy and test macro F1 exposes this effect clearly. Gradient Boosting achieves the highest test accuracy (0.880) but a test macro F1 of only 0.531 - a gap of 0.349 points, meaning the model is classifying baseline reliably but missing nearly half of all cognitive state events. Decision Tree is the most extreme overfitter, with perfect training scores (1.000) collapsing to a test macro F1 of 0.573. No model on raw time-domain features exceeds a test macro F1 of 0.55, establishing the ceiling that frequency-domain feature engineering must improve upon. It should be noted that these within-session scores are likely further inflated by temporal autocorrelation: at 256 Hz sampling, adjacent timesteps are nearly identical, so a random train/test split inadvertently places near-duplicate samples on both sides of the split.

Table 1: Benchmark classifier results on the simplified 4-pilot dataset, raw time-domain features (stratified split, 70/30). Macro F1 is the primary metric; the gap between test accuracy and test macro F1 quantifies how much each model exploits the majority baseline class.

| **Model** | **Tr. Acc.** | **Te. Acc.** | **Tr. Prec.** | **Te. Prec.** | **Tr. Rec.** | **Te. Rec.** | **Tr. F1** | **Te. F1** |
|---|---|---|---|---|---|---|---|---|
| KNN | 0.893 | 0.863 | 0.844 | 0.685 | 0.585 | 0.520 | 0.638 | 0.547 |
| Naive Bayes | 0.550 | 0.557 | 0.395 | 0.400 | 0.437 | 0.426 | 0.375 | 0.383 |
| Decision Tree | 1.000 | 0.827 | 1.000 | 0.569 | 1.000 | 0.578 | 1.000 | 0.573 |
| SVM (Linear SGD) | 0.763 | 0.767 | 0.382 | 0.384 | 0.402 | 0.405 | 0.391 | 0.394 |
| Neural Network | 0.836 | 0.830 | 0.669 | 0.416 | 0.447 | 0.444 | 0.435 | 0.430 |
| Gradient Boosting | 1.000 | 0.880 | 1.000 | 0.721 | 1.000 | 0.511 | 1.000 | 0.531 |

Table 2 shows the same models retrained on frequency-domain band power features and evaluated under a 5-fold stratified shuffle split. The improvement in macro F1 is substantial across all models. XGBoost achieves the highest mean macro F1 of 0.816 ($\sigma$ = 0.009), closely followed by LightGBM at 0.809 ($\sigma$ = 0.008). Both gradient boosted models also achieve test accuracy above 0.94, confirming that the frequency transformation improves performance on all classes simultaneously, not just the minority ones. Random Forest lags at 0.649, and the remaining models fall below 0.5 - MLP, KNN, and SVM benefit far less from the frequency-domain representation. The low standard deviation across folds for all models ($\leq$ 0.022) confirms these results are stable and not driven by a single favourable split. These within-session scores remain optimistic relative to cross-subject generalisation for the same reason noted above - windows from the same pilot appear on both sides of the split - but they establish the upper bound of what is achievable when generalization to unknown pilots is not a concern in model development.

Table 2: Within-session performance on the frequency-domain transformed dataset (5-fold stratified shuffle split, 70/30). All metrics are macro-averaged. Std F1 is the standard deviation of macro F1 across the 5 folds.

| **Model** | **Mean Acc.** | **Mean Prec.** | **Mean Rec.** | **Mean F1** | **Std F1** |
|---|---|---|---|---|---|
| XGBoost | 0.945 | 0.885 | 0.772 | 0.816 | 0.009 |
| LightGBM | 0.946 | 0.893 | 0.760 | 0.809 | 0.008 |
| Random Forest | 0.851 | 0.635 | 0.719 | 0.649 | 0.011 |
| MLP | 0.828 | 0.637 | 0.458 | 0.463 | 0.022 |
| KNN | 0.761 | 0.622 | 0.424 | 0.440 | 0.011 |
| SVM (Linear) | 0.681 | 0.405 | 0.348 | 0.344 | 0.003 |

Table 3 shows pilot-out cross-validation results on the 18-pilot frequency-domain dataset - the operationally relevant evaluation where the model is tested on a pilot it has never seen. XGBoost and LightGBM again lead with mean macro F1 of approximately 0.30, compared to 0.816–0.809 in the within-session evaluation. This drop of over 0.5 F1 points is attributable to between-pilot EEG variability and is discussed further in the next section.

Table 3: Pilot-out cross-validation on the 18-pilot frequency-domain dataset (18 folds, macro F1).

| **Model** | **Mean F1** | **Std** | **Min** | **Max** |
|---|---|---|---|---|
| XGBoost | 0.304 | 0.079 | 0.147 | 0.445 |
| LightGBM | 0.301 | 0.079 | 0.145 | 0.447 |
| MLP | 0.271 | 0.044 | 0.187 | 0.337 |
| Random Forest | 0.266 | 0.090 | 0.108 | 0.471 |
| KNN | 0.253 | 0.040 | 0.184 | 0.342 |
| SVM Linear ($\beta/\gamma$) | 0.237 | 0.055 | 0.187 | 0.395 |
| SVM Linear | 0.206 | 0.012 | 0.186 | 0.231 |

# 4 Discussion

## 4.1 Selecting a Suitable Performance Metric

The benchmark results in Table 1 illustrate why accuracy cannot be the primary selection criterion for a safety-critical classifier. With baseline events (events A and C) comprising over 85% of the data, a trivial model that always predicts "no alarming event" achieves high accuracy while detecting nothing useful. In an aviation safety context, a missed detection - failing to alert a pilot who is dangerously distracted - is far more costly than a false alarm. Macro F1 is the operationally appropriate metric because it penalizes a model equally for any class it fails on, regardless of how rarely that class appears. Under this criterion, Gradient Boosting's apparent superiority (test accuracy 0.880) is reduced by its test recall of 0.511 - half of all cognitive state events are going undetected.

## 4.2 The Role of Feature Engineering

The frequency-domain transformation is the most impactful methodological contribution of this work. Raw EEG amplitude has near-zero class separability across all channels (Figure 3), meaning that standard classifiers operating on the original features are effectively working with noise for the EEG portion of the dataset. Inferences based on data in this domain are almost exclusively based on ecg (heart rate) and gsr (essentially sweat), making this a less sophisticated classifier than this project sets out to make. Welch's method converts the inconsequential EEG signal amplitudes into beta and gamma band power features that carry some discriminative information (Figure 4), raising LightGBM's within-session macro F1 from the range achievable by the benchmark models to 0.848. Beta band activity is physiologically associated with active cognitive engagement and attentional switching, making this a theoretically grounded as well as empirically effective choice.

## 4.3 Best Performing Model and Method

On the within-session evaluation using frequency-domain features, XGBoost achieves the highest mean macro F1 of 0.816 ($\sigma = 0.009$), closely followed by LightGBM at 0.809 ($\sigma = 0.008$). Both models are stable across folds and substantially outperform all other classifiers, with Random Forest the next best at 0.649 and everything else falling below 0.5. The consistent advantage of gradient boosted trees over SVM, KNN, and MLP reflects their ability to learn complex non-linear interactions across the 89-feature frequency-domain space while naturally handling class imbalance through sample weighting. They are also robust to the mix of informative and uninformative features in the dataset, effectively ignoring the low-separability theta and alpha channels without requiring manual pruning.

On the pilot-out cross-validation across 18 subjects, XGBoost again leads with a mean macro F1 of 0.304 ($\sigma = 0.079$), with LightGBM at 0.301 ($\sigma = 0.079$). The relative ordering of models is consistent with the within-session results - gradient boosted trees outperform all alternatives - but the absolute performance drops dramatically for every model. The standard deviation of 0.079 across 18 folds is large relative to the mean, indicating that generalisation quality varies substantially depending on which pilot is held out. Some pilots are well-represented by the training distribution and yield fold scores above 0.44; others, whose signal profiles differ more from the training population, yield scores below 0.15. This fold-level variance is itself an important finding: it suggests that the system would perform reliably for some new pilots but unacceptably for others, which is precisely the kind of unpredictability that makes uncalibrated deployment unsafe.

XGBoost is therefore the recommended model for both evaluation settings. Its marginal edge over LightGBM is consistent across both within-session and pilot-out evaluation, and its handling of class imbalance via sample weighting is well-suited to the severe skew toward baseline events in this dataset.

## 4.4 The Cross-Subject Generalisation Problem

The performance drop from 0.848 within-session to 0.304 under pilot-out evaluation is the central finding of this work. Pilot identity consistently explains more variance in EEG band powers than cognitive state does. A classifier trained on known pilots inevitably learns individual physiological signatures that do not transfer to new individuals. The 4-pilot to 18-pilot ANOVA comparison further reveals that much of the

apparent discriminability in the small dataset was not real - driven by individual differences that happened to correlate with class labels in that particular sample, and not present in a more diverse population.

A calibration experiment demonstrated that adding even 5–20% of a new pilot's labelled data to the training set improves macro F1 from 0.280 to 0.344, with diminishing returns beyond 20%. In an operational context, this corresponds to a short pre-flight calibration session - feasible for a cockpit monitoring system and likely a practical necessity for acceptable performance on new individuals.

# 5 Conclusion

This study developed and systematically evaluated machine learning classifiers for real-time pilot cognitive state detection from EEG and physiological signals. The following conclusions can be drawn.

The frequency-domain transformation via Welch's method is essential: raw EEG amplitude carries near-zero class separability, while beta and gamma band powers provide the signal that gradient boosted classifiers can exploit. The best performing model overall is XGBoost trained on frequency-domain features, achieving macro F1 of 0.816 within session. Under the more rigorous and deployment-relevant pilot-out evaluation across 18 subjects, XGBoost leads at mean macro F1 of 0.304.

The gap between these two numbers - 0.816 versus 0.304 - is the defining result. It quantifies how much performance degrades when moving from a calibrated to an uncalibrated deployment scenario, and the variability analysis identifies the mechanism: EEG band powers encode pilot identity more strongly than cognitive state. Solving this could require either a substantially larger and more diverse pilot population or individual calibration data collected prior to each session. A short calibration session is the most immediately practical path, providing meaningful improvement with a small labeling cost.

The broader implication for aviation safety is that EEG-based cognitive monitoring is technically feasible within a session, but the deployment of a general monitoring tool requires careful consideration. Without calibration, the system will achieve far lower sensitivity to the dangerous states - DA and SS - that it is most important to detect.